

Data Sourcing & Licensing Analysis

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Project Context: Individual Study Plan (IPS) / Data Aggregator Application

Project Overview & Architectural Strategy

This document presents an analysis of the technical feasibility, data sourcing strategy, and licensing considerations for the proposed Context-Aware Data Aggregator application. Developed as an alternative project for the "Embedded Systems" course under an Individual Study Plan (IPS), the main goal is to design a system that securely collects, processes, and displays real-time environmental, meteorological, and geographic data tailored to the user's exact location.

1 Environmental & Weather Data

Open-Meteo

- **Purpose:** To retrieve precise pollen count data (via the Air Quality API) along with current wind speed, humidity, and temperature.
- **Licensing:** Provided under the Creative Commons Attribution 4.0 International (CC BY 4.0) license for non-commercial use. The service allows up to 10,000 daily API calls for free, which is more than sufficient for an educational project. The drawback is that it is free only for non-commercial purposes.
- **License Link:** [Open-Meteo Terms of Use](#)

OpenWeatherMap (Secondary Source)

- **Purpose:** To serve as a fallback or parallel source for general weather metrics, allowing us to limit requests to Open-Meteo if needed.
- **Licensing:** The self-service API is provided under the Open Database License (ODbL).
- **License Link:** [OpenWeatherMap Licensing Guide](#)

2 Geographic & Locality Context

BigDataCloud

- **Purpose:** To perform reverse geocoding to translate raw coordinates into human-readable administrative boundaries (e.g., determining if the location is a specific city, town, or village).
- **Licensing:** The client-side reverse geocoding API is completely free for both commercial and non-commercial use without throttling. The JavaScript client itself is open-source under the MIT License.
- **License Link:** [BigDataCloud Reverse Geocoding GitHub Repository \(MIT License\)](#)

OpenStreetMap (Overpass API)

- **Purpose:** To query specific geographic features around the user's coordinates, specifically to determine proximity to water (waterways, coastlines) and classify the surrounding land use (e.g., rural vs. urban).
- **Licensing:** OpenStreetMap data is available under the Open Database License (ODbL) 1.0. It allows users to copy, distribute, and adapt the data as long as OpenStreetMap contributors are properly credited.
- **License Link:** [OpenStreetMap Copyright and License](#)

3 System & Astronomy Data

SunriseSunset.io

- **Purpose:** To calculate exact sunrise and sunset times dynamically by passing the current latitude, longitude, and date.
- **Licensing:** The API is publicly available and free to use without an API key, requiring only standard attribution.
- **License Link:** [SunriseSunset.io API Terms](#)

Native Device APIs

- **Purpose:** To securely access the user's current coordinates and system time.
- **Licensing:** The project will utilize the native Geolocation API and the JavaScript `Date()` object. These are built-in platform standards and do not require external commercial licensing.